Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Theme 3: KoboToolbox

Self-Hosted Installation, Configuration & CRD Integration

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Day 3 of 4 | crd.resilienceacademy.ac.tz · Use case: Tanga Flood Survey

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Theme 3 Overview

Goal: Install a self-hosted KoboToolbox and pipe survey data into CRD

Morning

- KoboToolbox use cases & RA context
- Architecture and hosting options
- Practical 3.1: Installation (3h)
- Practical 3.2: Form design – Tanga flood survey (2h)

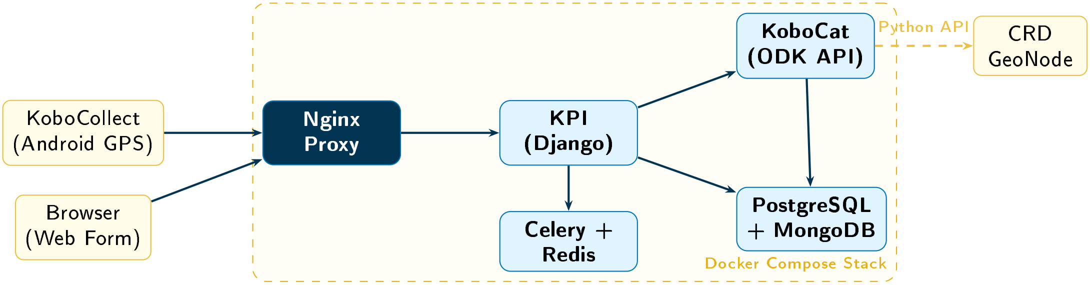
Afternoon

- Practical 3.3: Backend maintenance (2h)
- Practical 3.4: KoboToolbox to CRD integration (1h)
- Q&A and troubleshooting

Context: Resilience Academy uses KoboToolbox with GIZ partners for Tanga flood mapping

Alumni collect GPS flood data in the field → data flows into CRD

KoboToolbox Architecture



Technology Stack

React frontend · Django/KPI backend · PyXForm/ODK engine · PostgreSQL + MongoDB · Nginx · Docker

KoboToolbox Installation

Steps 1-2: Clone + Wizard

```
git clone https://github.com/kobotoolbox/
  kobo-install.git
cd kobo-install

# Run interactive setup wizard
python3 run.py
# Answer:
# Installation type? -> Local
# Local domain? -> localhost
# Use HTTPS? -> No
# Advanced options? -> No
```

Steps 3-4: Start + Verify

```
# Start all services
python3 run.py --start

# Watch startup (5-10 minutes)
docker compose logs -f kpi
# Wait for:
# Listening at http://0.0.0.0:8000

# Create admin account
docker compose exec kpi \
  python manage.py createsuperuser

# Verify all services
docker compose ps
# Access: http://localhost
```

Tanga Flood Survey Form Design

Form Questions (XLSForm)

Question	Type
GPS Location	Point (geopoint)
Flood depth (m)	Decimal number
Date observed	Date
Ward name	Text
Photo of site	Image
Affected households	Integer
Water source	Select one
Infrastructure damage	Select multiple
Observer notes	Text area

Why GPS matters

GPS questions auto-generate **point geometries** which export directly as GeoJSON for upload to CRD.

Integration Flow

KoboCollect GPS Survey → REST API JSON → Python GeoJSON → CRD Layer

Hosting Options & Data Sovereignty

Cloud (UNHCR)

`kobo.humanitarian response.info`

Free. No setup. Data on external servers. Limited control.

Self-Hosted

Our choice.

Full data control. Required for Tanga/Mwanza government data. Runs on your own server.

KoboToolbox.org

`kf.kobotoolbox.org`

Free tier. UNHCR managed. Good for small projects.

RA/GIZ Tanga Flood Mapping Context

Tanga flood survey data collected by RA alumni must remain under institutional control. Self-hosting at RA infrastructure ensures data sovereignty and compliance with national data governance requirements.

Day 3 Summary

Achieved

- ✓ KoboToolbox architecture and hosting options
- ✓ Self-hosted server installed with kobo-install
- ✓ Tanga flood survey form with GPS deployed
- ✓ Backend maintenance: backups, updates, users
- ✓ Tanga flood data piped into CRD via Python API

Day 4: Integration, Basemaps & APIs

- CRD as WMS basemap in Leaflet
- Tanga and Mwanza layers in web app
- Link CRD to QGIS and Python
- CRD REST API statistics dashboard
- Wrap-up and certificates

Bring your running CRD instance!

Q&A and Troubleshooting

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